Hein Zarni Aung

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EDUCATION

Stony Brook University **Expected Graduation: May 2027**

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Relevant Coursework: Algorithms, Statistics & Probability, Systems Fundamentals I & II

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University **Sep 2025 – Dec 2025**

- Built a Python pipeline using Selenium to monitor newly issued SSL/TLS certificates in real time, applying keyword- and heuristic-based detection logic to flag fraudulent domains impersonating EZ-Pass tolling services.

Vertically Integrated Project — Automotive Ethics Lab, Stony Brook University **Starting Fall 2026**

Joining the Simulations sub-team to build driving/traffic scenario simulations studying how autonomous systems should make ethical decisions, a multi-semester commitment through graduation.

INTERNSHIPS

Software Engineering Intern — Clustr **Jan 2026 – Sept 2026 (expected)**

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation.

Software Engineering Intern — Prenora Dynamics LLC **Jun 2026 – Sept 2026 (expected)**

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

PROJECTS

Diary Handwriting Analysis **Python, PyTorch, Hugging Face Transformers, OpenCV**

- Fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline (grayscale conversion, binary thresholding, horizontal ink-projection profiling), reaching ~87% reliable page transcription on personal diary pages.
- Pivoted to DANIEL (Document Attention Network for Information Extraction and Labeling), a segmentation-free architecture that transcribes full pages directly and jointly extracts named entities, to enable topic and mood-trend statistics across diary entries over time; currently in progress.

TuneAcademy **React, NumPy, Essentia, Firestore**

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on specific technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Evaluated a planned machine learning model against rule-based scoring algorithms under hackathon time and compute constraints, choosing the rule-based approach to deliver reliable weakness detection within the 36-hour build window.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University **Aug 2025 – Dec 2025**

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original PowerPoint-based review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

Machine Learning: PyTorch, Hugging Face Transformers, OpenCV, NumPy

AI Tools: Claude Code

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP, Supabase

Tools: Git, Docker, Linux, Selenium